

Neuron structure and organization reflect function in information transfer

Our starting point for exploring the nervous system is the **neuron**, a cell type exemplifying the close fit of form and function that often arises over the course of evolution.

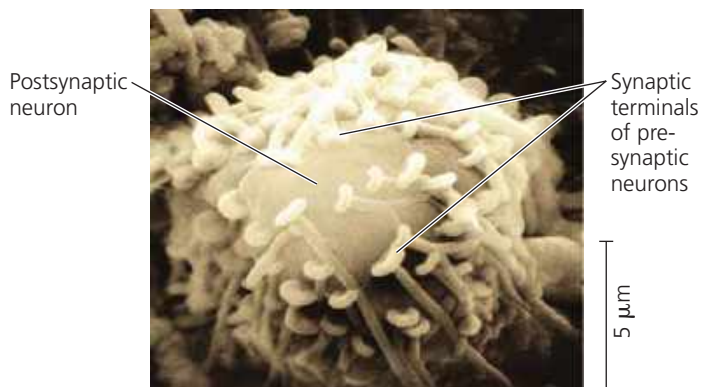
Neuron Structure and Function

The ability of a neuron to receive and transmit information is based on the highly specialized cellular organization shown in Figure 48.1. Most of a neuron's organelles, including its nucleus, are located in the **cell body**. In a typical neuron, the cell body is studded with numerous highly branched extensions called **dendrites** (from the Greek *dendron*, tree). Together with the cell body, the dendrites *receive* signals from other neurons.

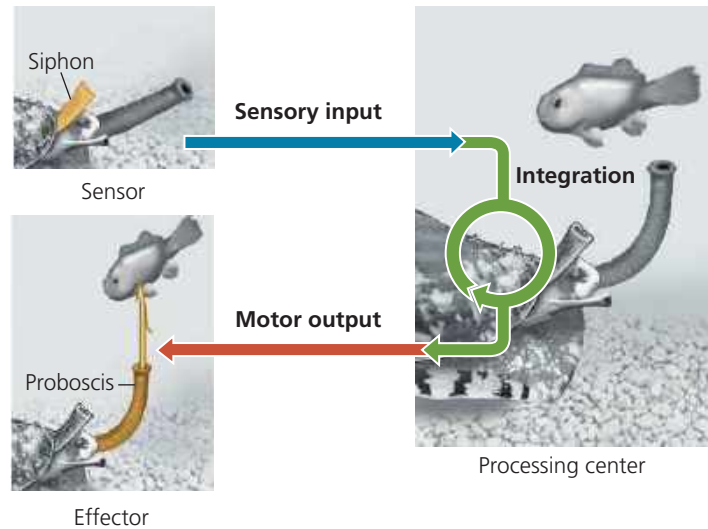
A typical neuron has a single **axon**, the extension that *transmits* signals to other cells. Axons are often much longer than dendrites, and some, such as those that reach from the spinal cord of a giraffe to the muscle cells in its feet, are over a meter long. The specialized structure of axons allows them to use pulses of electrical current to transmit information, even over long distances. The cone-shaped base of an axon, called the axon hillock, is typically where signals that travel down the axon are generated. Near its other end, an axon usually divides into many branches.

Each branched end of an axon transmits information to another cell at a junction called a **synapse** (Figure 48.2). The part of each axon branch that forms this specialized junction is a *synaptic terminal*. At most synapses, chemical messengers called **neurotransmitters** pass information from the transmitting neuron to the receiving cell. Cone snail venom is particularly potent because it interferes not only with electrical signaling along axons but also with chemical signaling across synapses.

▼ **Figure 48.2** Synaptic terminals on the cell body of a postsynaptic neuron. (colored SEM)



▼ **Figure 48.3** Summary of information processing. The cone snail's siphon acts as a sensor, transferring information to the neuronal circuits in the snail's head for processing. If prey is detected, these circuits issue motor commands—signals that control muscle activity. In this example, motor commands trigger release of a harpoon-like tooth from the proboscis, spearing the prey.



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Introduction to Information Processing

Information processing by a nervous system occurs in three stages: sensory input, integration, and motor output. As an example, let's consider the cone snail, focusing on the steps involved in identifying and attacking its prey. To generate sensory input to the nervous system, the snail surveys its environment with sensors in its tubelike siphon, sampling scents that might reveal a nearby fish (Figure 48.3). During the integration stage, networks of neurons in the snail brain process this information to determine if a fish is in fact present and, if so, where the fish is located. Motor output from the processing center then initiates attack, activating neurons that trigger release of the harpoon-like tooth toward the prey.

In all but the simplest animals, specialized populations of neurons handle each stage of information processing.

- **Sensory neurons**, like those in the snail's siphon, transmit information about external stimuli (such as light, touch, or smell), and internal conditions (such as blood pressure or muscle tension).
- **Interneurons** form the local circuits connecting neurons in the brain or ganglia. Interneurons are responsible for the integration (analysis and interpretation) of sensory input.
- **Motor neurons** transmit signals to muscle cells, causing them to contract. Additional neurons that extend out of the processing centers trigger gland activity.

All neurons transmit electrical signals within the cell in an identical manner. Thus a neuron that detects an odor transmits